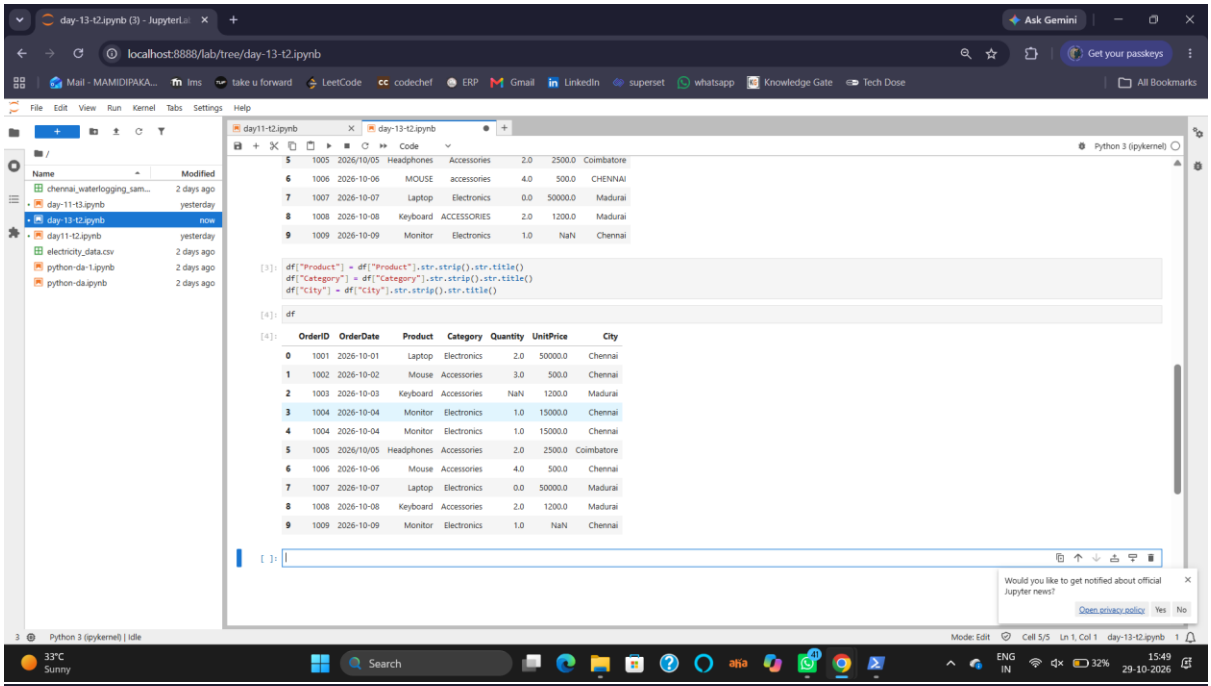
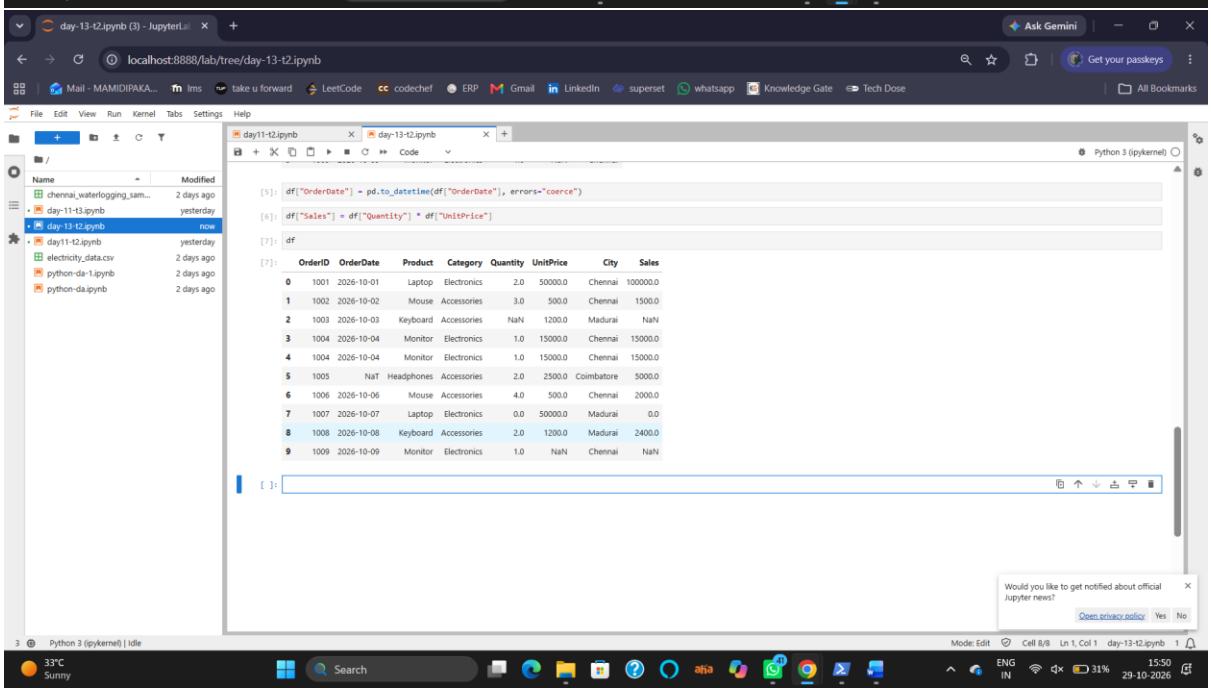
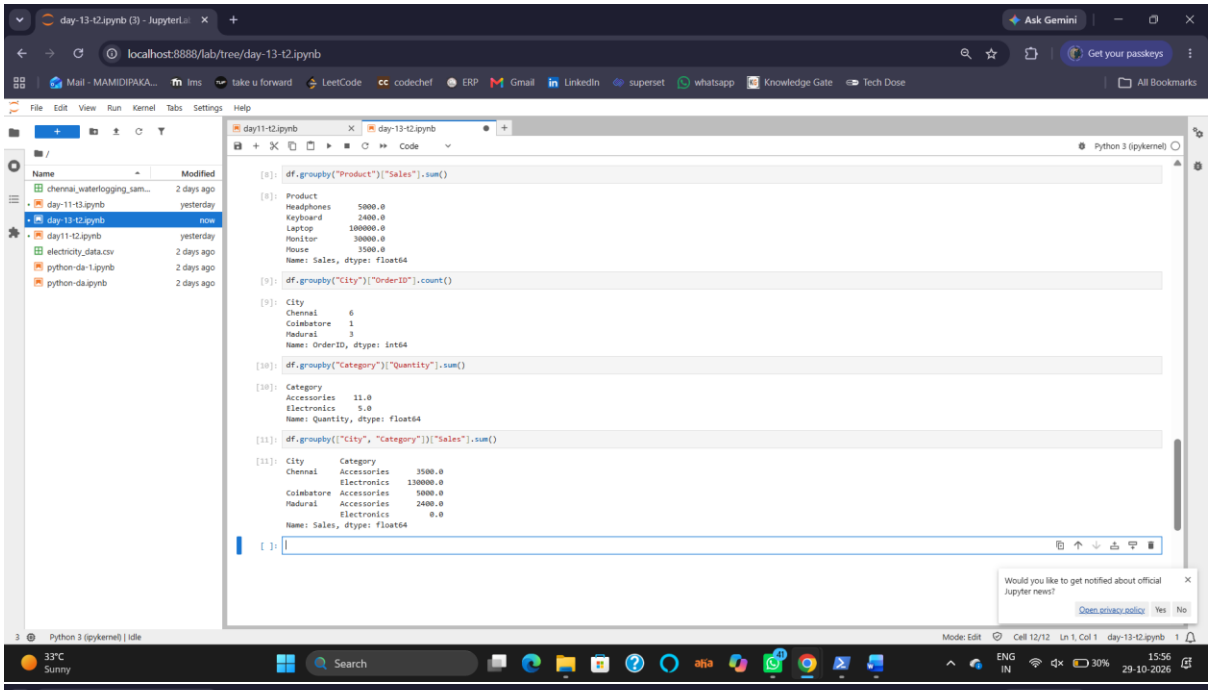
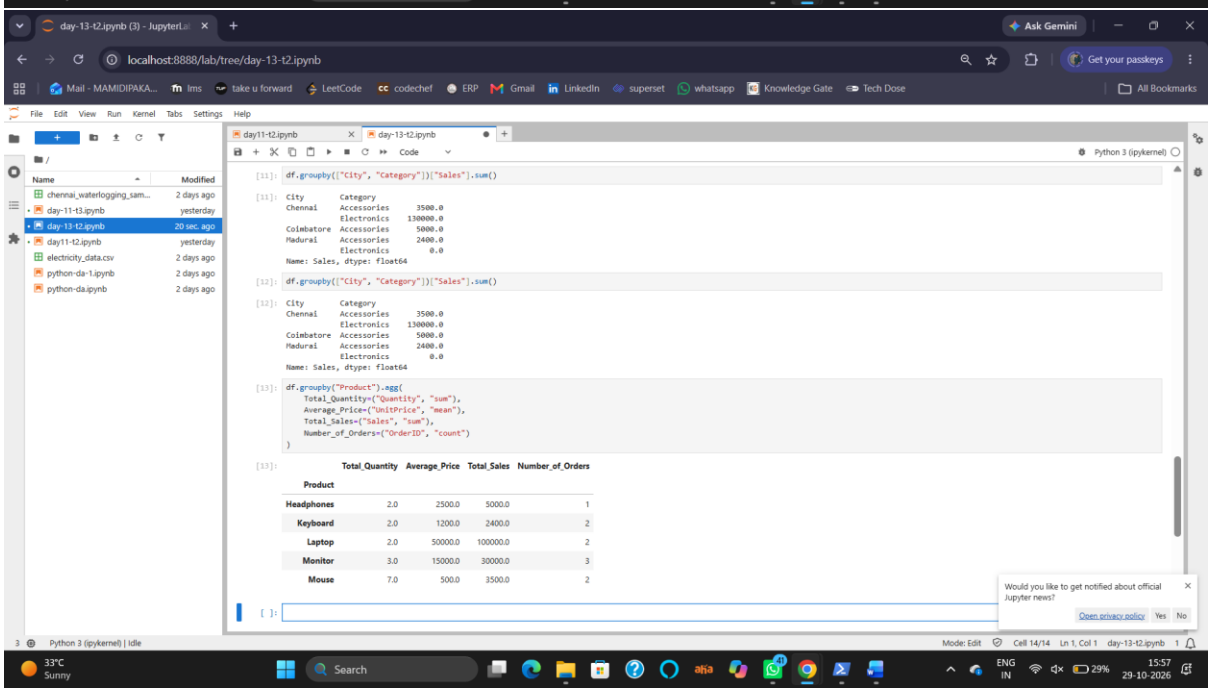
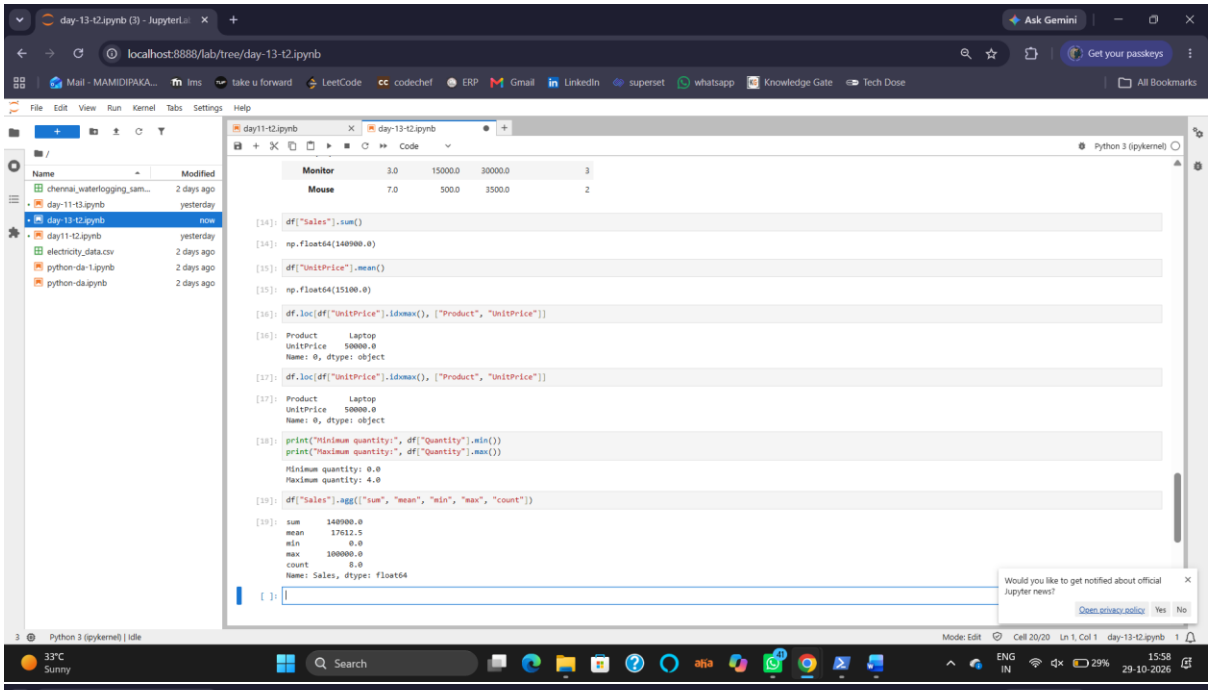
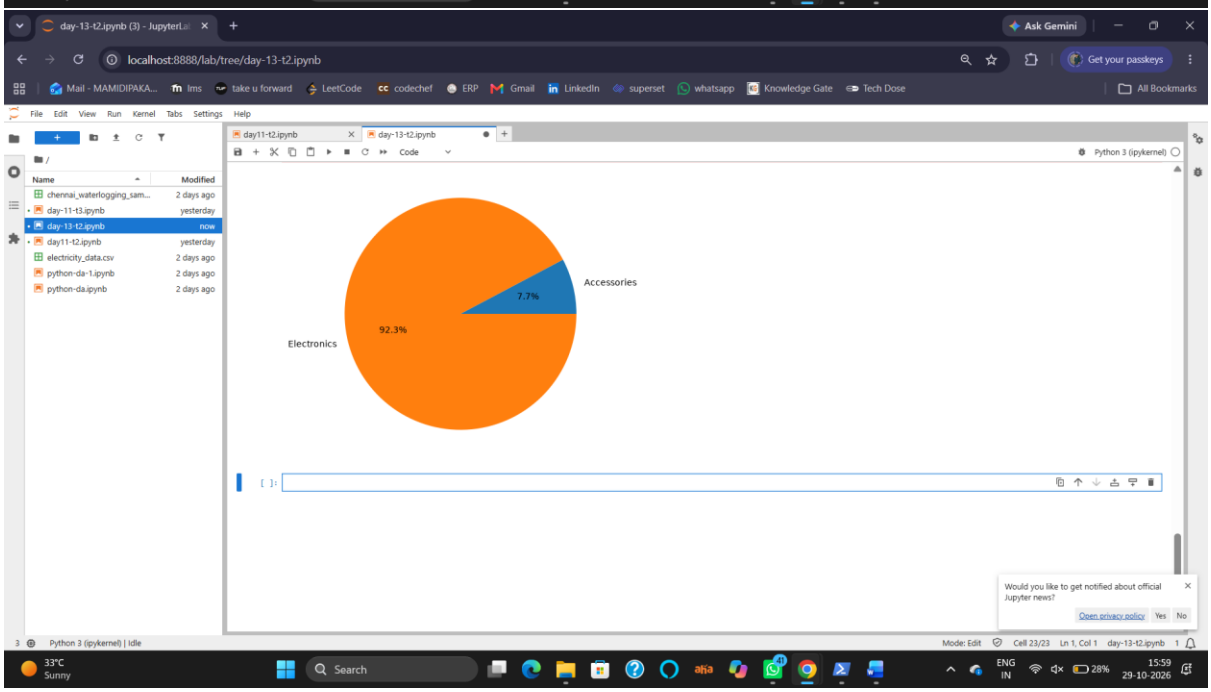

day-13-t2.ipynb (3) - JupyterLab
localhost:8888/lab/tree/day-13-t2.ipynb
Ask Gemini
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlogging_sam... 2 days ago
day-11-t3.ipynb yesterday
day-13-t2.ipynb now
day11-t2.ipynb yesterday
electricity_data.csv 2 days ago
python-da-1.ipynb 2 days ago
python-da.ipynb 2 days ago
5 1005 2026/10/05 Headphones Accessories 2.0 2500.0 Coimbatore
6 1006 2026-10-06 MOUSE accessories 4.0 500.0 CHENNAI
7 1007 2026-10-07 Laptop Electronics 0.0 50000.0 Madurai
8 1008 2026-10-08 Keyboard ACCESSORIES 2.0 1200.0 Madurai
9 1009 2026-10-09 Monitor Electronics 1.0 NaN Chennai
[3]: df["Product"] = df["Product"].str.strip().str.title()
df["Category"] = df["Category"].str.strip().str.title()
df["City"] = df["City"].str.strip().str.title()
[4]: df
[4]:
OrderID OrderDate Product Category Quantity UnitPrice City
0 1001 2026-10-01 Laptop Electronics 2.0 50000.0 Chennai
1 1002 2026-10-02 Mouse Accessories 3.0 500.0 Chennai
2 1003 2026-10-03 Keyboard Accessories NaN 1200.0 Madurai
3 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai
4 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai
5 1005 2026/10/05 Headphones Accessories 2.0 2500.0 Coimbatore
6 1006 2026-10-06 Mouse Accessories 4.0 500.0 Chennai
7 1007 2026-10-07 Laptop Electronics 0.0 50000.0 Madurai
8 1008 2026-10-08 Keyboard Accessories 2.0 1200.0 Madurai
9 1009 2026-10-09 Monitor Electronics 1.0 NaN Chennai
[ ]:
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 5/5 Ln 1, Col 1 day-13-t2.ipynb
33°C Sunny
Search
ENG IN
32%
15:49 29-10-2026

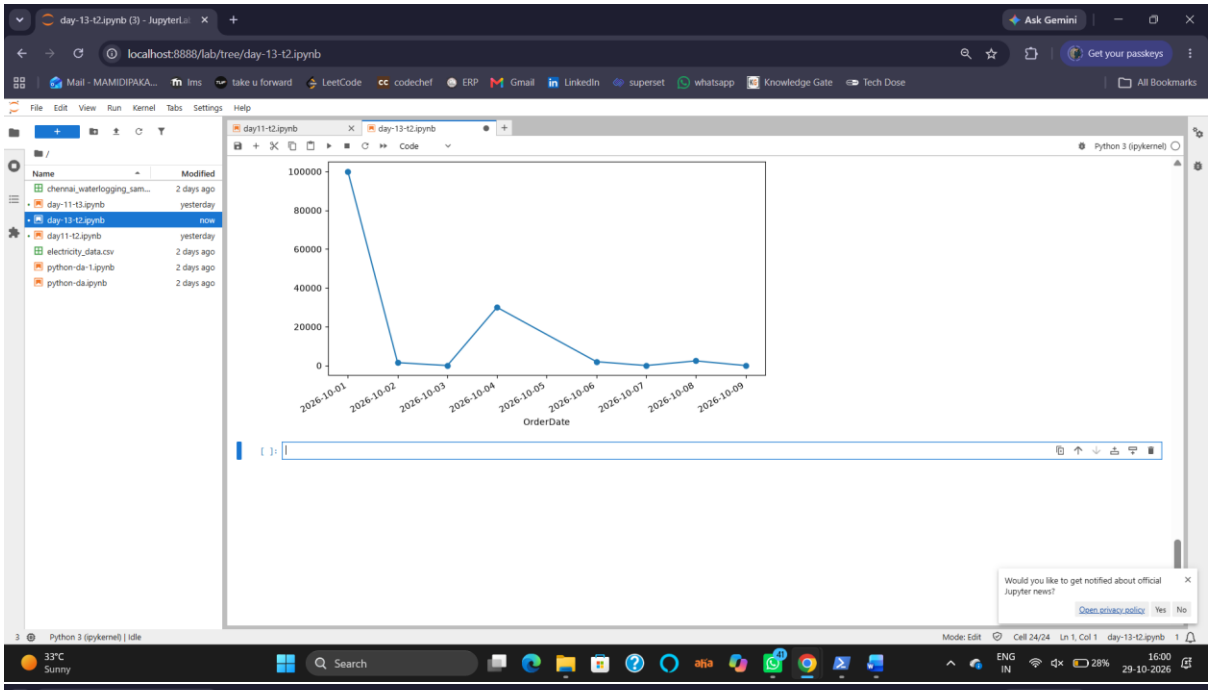

day-13-t2.ipynb (3) - JupyterLab
localhost:8888/lab/tree/day-13-t2.ipynb
Ask Gemini
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlogging_sam... 2 days ago
day-11-t3.ipynb yesterday
day-13-t2.ipynb now
day11-t2.ipynb yesterday
electricity_data.csv 2 days ago
python-da-1.ipynb 2 days ago
python-da.ipynb 2 days ago
[5]: df["OrderDate"] = pd.to_datetime(df["OrderDate"], errors="coerce")
[6]: df["Sales"] = df["Quantity"] * df["UnitPrice"]
[7]: df
[7]:
OrderID OrderDate Product Category Quantity UnitPrice City Sales
0 1001 2026-10-01 Laptop Electronics 2.0 50000.0 Chennai 100000.0
1 1002 2026-10-02 Mouse Accessories 3.0 500.0 Chennai 1500.0
2 1003 2026-10-03 Keyboard Accessories NaN 1200.0 Madurai NaN
3 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai 15000.0
4 1004 2026-10-04 Monitor Electronics 1.0 15000.0 Chennai 15000.0
5 1005 NaT Headphones Accessories 2.0 2500.0 Coimbatore 5000.0
6 1006 2026-10-06 Mouse Accessories 4.0 500.0 Chennai 2000.0
7 1007 2026-10-07 Laptop Electronics 0.0 50000.0 Madurai 0.0
8 1008 2026-10-08 Keyboard Accessories 2.0 1200.0 Madurai 2400.0
9 1009 2026-10-09 Monitor Electronics 1.0 NaN Chennai NaN
[ ]:
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 8/8 Ln 1, Col 1 day-13-t2.ipynb
33°C Sunny
Search
ENG IN
31%
15:50 29-10-2026

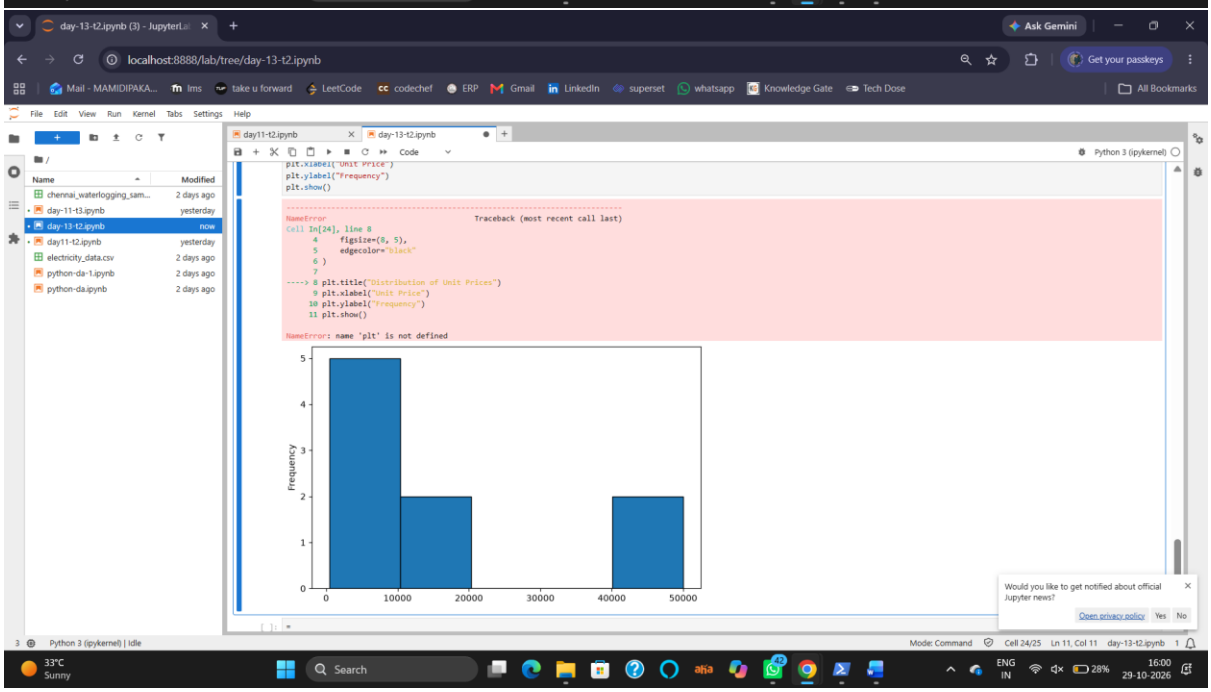

day-13-t2.ipynb (3) - JupyterLab
localhost:8888/lab/tree/day-13-t2.ipynb
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
[8]: df.groupby("Product")["Sales"].sum()
Product
Headphones 5000.0
Keyboard 2400.0
Laptop 100000.0
Monitor 30000.0
Mouse 3500.0
Name: Sales, dtype: float64
[9]: df.groupby("City")["OrderID"].count()
City
Chennai 6
Coimbatore 1
Madurai 3
Name: OrderID, dtype: int64
[10]: df.groupby("Category")["Quantity"].sum()
Category
Accessories 11.0
Electronics 5.0
Name: Quantity, dtype: float64
[11]: df.groupby(["City", "Category"])["Sales"].sum()
City Category
Chennai Accessories 3500.0
Electronics 130000.0
Coimbatore Accessories 5000.0
Madurai Accessories 2400.0
Electronics 0.0
Name: Sales, dtype: float64
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 12/12 Ln 1, Col 1 day-13-t2.ipynb
33°C Sunny
Search
ENG IN
30%
15:56 29-10-2026

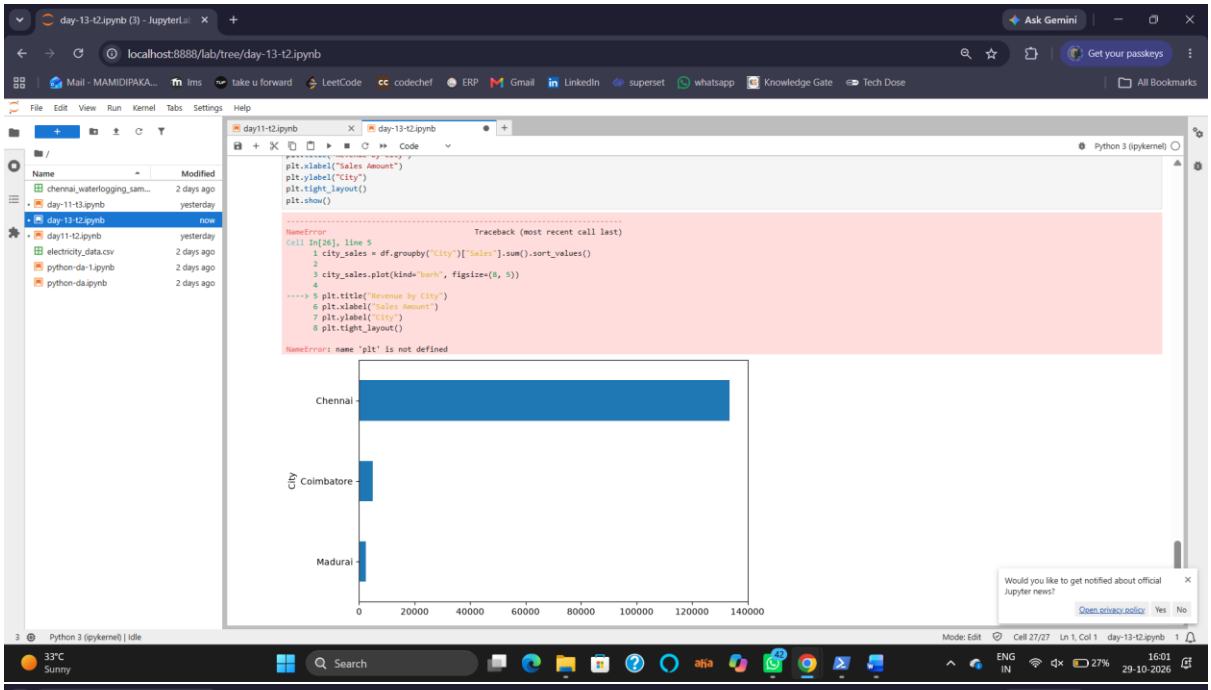

day-13-t2.ipynb (3) - JupyterLab
localhost:8888/lab/tree/day-13-t2.ipynb
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
[11]: df.groupby(["City", "Category"])["Sales"].sum()
City Category
Chennai Accessories 3500.0
Electronics 130000.0
Coimbatore Accessories 5000.0
Madurai Accessories 2400.0
Electronics 0.0
Name: Sales, dtype: float64
[12]: df.groupby(["City", "Category"])["Sales"].sum()
[13]: df.groupby("Product").agg(
Total_Quantity=("Quantity", "sum"),
Average_Price=("UnitPrice", "mean"),
Total_Sales=("Sales", "sum"),
Number_of_Orders=("OrderID", "count")
)
Total_Quantity Average_Price Total_Sales Number_of_Orders
Product
Headphones 2.0 2500.0 5000.0 1
Keyboard 2.0 1200.0 2400.0 2
Laptop 2.0 50000.0 100000.0 2
Monitor 3.0 15000.0 30000.0 3
Mouse 7.0 500.0 3500.0 2
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 14/14 Ln 1, Col 1 day-13-t2.ipynb
33°C Sunny
Search
ENG IN
29%
15:57 29-10-2026

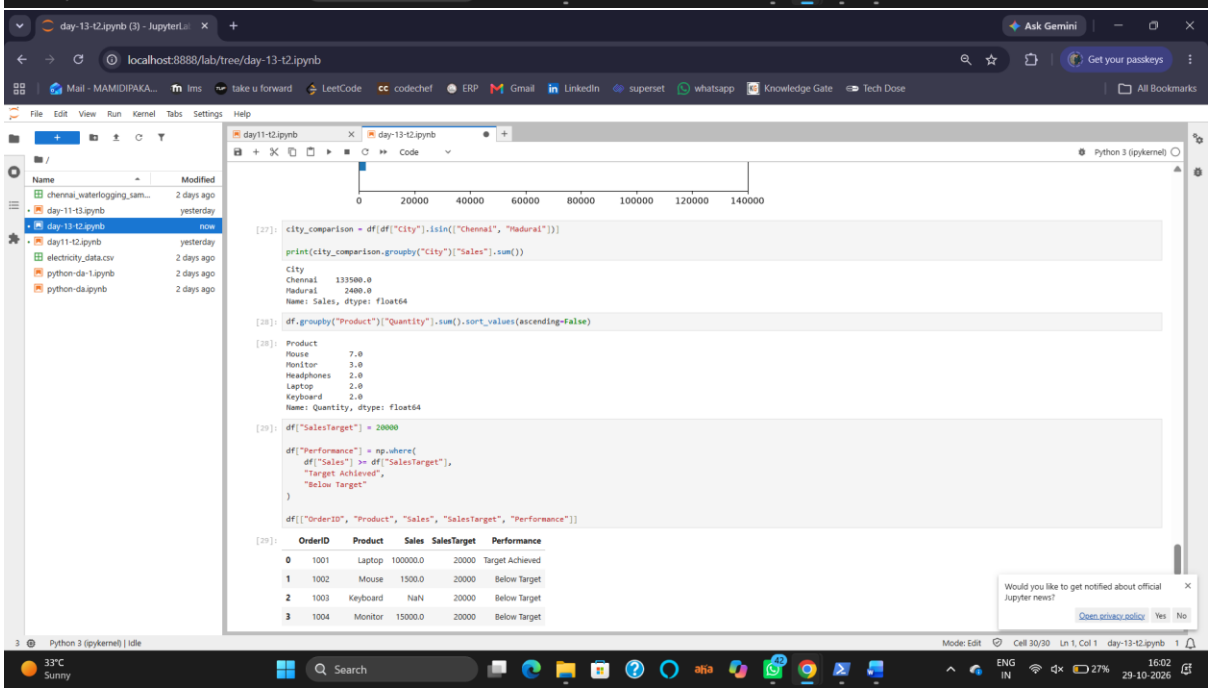

day-13-t2.ipynb (3) - JupyterLab
Ask Gemini
localhost:8888/lab/tree/day-13-t2.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
Monitor 3.0 15000.0 30000.0 3
Mouse 7.0 500.0 3500.0 2
[14]: df["Sales"].sum()
[14]: np.float64(140900.0)
[15]: df["UnitPrice"].mean()
[15]: np.float64(15100.0)
[16]: df.loc[df["UnitPrice"].idxmax(), ["Product", "UnitPrice"]]
[16]: Product Laptop
UnitPrice 50000.0
Name: 0, dtype: object
[17]: df.loc[df["UnitPrice"].idxmax(), ["Product", "UnitPrice"]]
[17]: Product Laptop
UnitPrice 50000.0
Name: 0, dtype: object
[18]: print("Minimum quantity:", df["Quantity"].min())
print("Maximum quantity:", df["Quantity"].max())
Minimum quantity: 0.0
Maximum quantity: 4.0
[19]: df["Sales"].agg(["sum", "mean", "min", "max", "count"])
[19]: sum 140900.0
mean 17612.5
min 0.0
max 100000.0
count 8.0
Name: Sales, dtype: float64
[ ]:
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 20/20 Ln 1, Col 1 day-13-t2.ipynb
33°C Sunny
Search
ENG IN
29%
15:58 29-10-2026


day-13-t2.ipynb (3) - JupyterLab
Ask Gemini
localhost:8888/lab/tree/day-13-t2.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
Accessories
7.7%
92.3%
Electronics
[ ]:
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 23/23 Ln 1, Col 1 day-13-t2.ipynb
33°C Sunny
Search
ENG IN
28%
15:59 29-10-2026


day-13-t2.ipynb (3) - JupyterLab
localhost:8888/lab/tree/day-13-t2.ipynb
Ask Gemini
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
Name
Modified
chennai_waterlogging_sam...
day-11-t3.ipynb
day-13-t2.ipynb
day11-t2.ipynb
electricity_data.csv
python-da-1.ipynb
python-da.ipynb
Code
Python 3 (ipykernel)
100000
80000
60000
40000
20000
0
2026-10-01
2026-10-02
2026-10-03
2026-10-04
2026-10-05
2026-10-06
2026-10-07
2026-10-08
2026-10-09
OrderDate
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 24/24
Ln 1, Col 1
33°C
Sunny
Search
ENG
IN
28%
16:00
29-10-2026


day-13-t2.ipynb (3) - JupyterLab
localhost:8888/lab/tree/day-13-t2.ipynb
Ask Gemini
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
plt.xlabel("Unit Price")
plt.ylabel("Frequency")
plt.show()
NameError Traceback (most recent call last)
Cell In[24], line 8
4 figsize=(8, 5),
5 edgecolor="black"
6)
7
----> 8 plt.title("Distribution of Unit Prices")
9 plt.xlabel("Unit Price")
10 plt.ylabel("Frequency")
11 plt.show()
NameError: name 'plt' is not defined
Frequency
5
4
3
2
1
0
0
10000
20000
30000
40000
50000
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Command
Cell 24/25
Ln 11, Col 11
33°C
Sunny
Search
ENG
IN
28%
16:00
29-10-2026


day-13-t2.ipynb (3) - JupyterLa
Ask Gemini
localhost:8888/lab/tree/day-13-t2.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlogging_sam... 2 days ago
day-11-t3.ipynb yesterday
day-13-t2.ipynb now
day11-t2.ipynb yesterday
electricity_data.csv 2 days ago
python-da-1.ipynb 2 days ago
python-da.ipynb 2 days ago
plt.xlabel("Sales Amount")
plt.ylabel("City")
plt.tight_layout()
plt.show()
NameError Traceback (most recent call last)
Cell In[26], line 5
1 city_sales = df.groupby("City")["Sales"].sum().sort_values()
2
3 city_sales.plot(kind="barh", figsize=(8, 5))
4
----> 5 plt.title("Revenue by City")
6 plt.xlabel("Sales Amount")
7 plt.ylabel("City")
8 plt.tight_layout()
NameError: name 'plt' is not defined
Chennai
Coimbatore
Madurai
City
0
20000
40000
60000
80000
100000
120000
140000
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 27/27
Ln 1, Col 1
day-13-t2.ipynb
33°C
Sunny
Search
ENG
IN
27%
16:01
29-10-2026


day-13-t2.ipynb (3) - JupyterLa
Ask Gemini
localhost:8888/lab/tree/day-13-t2.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
day11-t2.ipynb
day-13-t2.ipynb
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlogging_sam... 2 days ago
day-11-t3.ipynb yesterday
day-13-t2.ipynb now
day11-t2.ipynb yesterday
electricity_data.csv 2 days ago
python-da-1.ipynb 2 days ago
python-da.ipynb 2 days ago
0
20000
40000
60000
80000
100000
120000
140000
[27]:
city_comparison = df[df["City"].isin(["Chennai", "Madurai"])]
print(city_comparison.groupby("City")["Sales"].sum())
City
Chennai 133500.0
Madurai 2400.0
Name: Sales, dtype: float64
[28]:
df.groupby("Product")["Quantity"].sum().sort_values(ascending=False)
[28]:
Product
Mouse 7.0
Monitor 3.0
Headphones 2.0
Laptop 2.0
Keyboard 2.0
Name: Quantity, dtype: float64
[29]:
df["SalesTarget"] = 20000
df["Performance"] = np.where(
df["Sales"] >= df["SalesTarget"],
"Target Achieved",
"Below Target"
)
df[["OrderID", "Product", "Sales", "SalesTarget", "Performance"]]
[29]:
OrderID Product Sales SalesTarget Performance
0 1001 Laptop 100000.0 20000 Target Achieved
1 1002 Mouse 1500.0 20000 Below Target
2 1003 Keyboard NaN 20000 Below Target
3 1004 Monitor 15000.0 20000 Below Target
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 30/30
Ln 1, Col 1
day-13-t2.ipynb
33°C
Sunny
Search
ENG
IN
27%
16:02
29-10-2026